ReliefPlan AI: Autonomous Humanitarian Camp Layout & Geospatial Optimization

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Abstract

Rapid deployment of humanitarian relief camps following natural disasters or armed conflicts remains a critical, time-sensitive challenge. Existing manual planning approaches -- dependent on expert field teams and printed handbooks -- frequently lead to overcrowding, inequitable access to water and sanitation, and suboptimal land utilisation. This paper presents **ReliefPlan AI**, an end-to-end AI-driven pipeline that accepts satellite or drone imagery of a disaster-affected site and automatically outputs a fully annotated, standards-compliant camp blueprint. The pipeline comprises five stages: (1) image acquisition; (2) terrain semantic segmentation using a custom DeepLabV3+ architecture with Atrous Spatial Pyramid Pooling (ASPP) at dilation rates r in {6, 12, 18}; (3) hydrological flood-buffer analysis; (4) a constraint-based spatial optimisation engine that places shelter blocks, hierarchical road networks, potable water tap stands, sanitation blocks, medical centres, food courts, and water purification reservoirs while satisfying UNHCR Emergency Settlement Standards; and (5) a geo-referenced blueprint renderer. Results show that the proposed DeepLabV3+ achieves mIoU **44.9%** and Dice **61.1%**, outperforming U-Net (mIoU 30.3%, Dice 46.1%) with 8x fewer parameters and 2.4x lower latency. The optimisation engine achieves a **99.4% UNHCR compliance score** and guarantees water and sanitation access in every residential block.

Keywords: *Disaster relief planning * DeepLabV3+ * ASPP * Semantic segmentation * UNHCR standards * Constraint optimisation * Humanitarian AI * Satellite imagery*

1. Introduction

Natural disasters -- including earthquakes, floods, tropical cyclones, and severe weather events -- alongside armed conflicts and political crises, displace millions of vulnerable people globally each year [1]. The rapid and methodical establishment of safe, sanitary, and highly accessible temporary shelter is widely recognised as the most urgent first-order humanitarian priority in the immediate aftermath of any large-scale crisis event. Delays or significant inefficiencies in deploying emergency camp infrastructure directly translate into excess morbidity and preventable mortality, particularly among vulnerable populations such as children and the elderly. Beyond immediate survival, the spatial configuration of these temporary settlements profoundly influences the quality of daily life, the probability of communicable disease outbreaks, and the efficiency of logistical supply chains for food and water.

Current industry practice for emergency camp design is characterised by severe limitations that impede the speed of humanitarian response. A small, frequently overburdened team of expert humanitarian planners is typically tasked with this complex design problem under extreme time pressure, often relying on rudimentary GIS desktop tools and printed field handbooks published by authoritative bodies such as UNHCR [2]. Because the design process is highly manual and iterative, the resulting layouts frequently exhibit avoidable inefficiencies: irregular road networks that impede ambulance access, shelter blocks inadvertently placed on flood-prone terrain, and sanitation facilities situated beyond the maximum recommended walkable distance, leading to localized health hazards and chronic water insecurity.

ReliefPlan AI addresses these critical, systemic limitations through a fully automated, end-to-end planning pipeline driven by deep learning and constraint satisfaction. Given only a raw optical satellite image of the prospective camp site and an estimated target population size, the system operates autonomously to produce a mathematically rigorous, standards-compliant solution in under 60 seconds. The pipeline comprises five stages: (i) terrain semantic segmentation using a custom DeepLabV3+ architecture with Atrous Spatial Pyramid Pooling (ASPP); (ii) hydrological flood-safety buffer enforcement; (iii) a deterministic, constraint-based spatial optimisation engine that rigorously packs shelter pods, road networks, water tap stands, communal latrines, medical centres, and food courts; (iv) high-resolution blueprint rendering; and (v) comprehensive compliance and logistics reporting. All generated layouts strictly satisfy UNHCR Emergency Settlement Standards, completely eliminating the need for manual CAD drafting.

2. Related Work

2.1 Semantic Segmentation of Overhead Imagery

The domain of semantic segmentation of overhead and satellite imagery has undergone remarkable progress over the past decade, driven primarily by advances in deep convolutional neural network (CNN) architectures. Long et al. [3] introduced Fully Convolutional Networks (FCNs) in 2015, establishing the seminal encoder-decoder paradigm. Building upon this, Ronneberger et al. [4] introduced symmetric skip connections to create the U-Net architecture, which rapidly became the baseline for remote-sensing land-cover classification due to its strong performance with scarce annotated training data. More recently, Chen et al. [5] proposed DeepLabV3+, which incorporates Atrous Spatial Pyramid Pooling (ASPP). ASPP applies parallel dilated convolutions to aggregate multi-scale semantic context without sacrificing spatial resolution. This capability is critically important for aerial imagery analysis, where buildings, roads, and water bodies span a wide and unpredictable range of spatial scales. Training robust models for these tasks relies heavily on public benchmarks such as LoveDA [6] and SpaceNet [7], which provide the geographic diversity necessary for humanitarian applications.

2.2 Automated Humanitarian Planning & Layout Optimization

The Facility Layout Problem (FLP) -- the problem of optimally arranging a set of interacting facilities within a bounded space to minimise operational costs, maximise accessibility, or satisfy a set of hard geometric constraints -- has been studied extensively in industrial engineering and operations research literature [10]. Recent advances have explored the application of Deep Reinforcement Learning (DRL) agents trained with Proximal Policy Optimisation (PPO) [11, 12] to solve variants of the FLP with promising results, demonstrating the potential of learned policies to generalise across novel layout configurations.

However, the application of AI-driven automated planning to humanitarian settlement design remains a critically underexplored frontier. UNHCR and the Sphere Project publish highly detailed, rigorously validated field operational standards [1, 2] specifying per-capita minimum area requirements, maximum acceptable walking distances to potable water, sanitation, and medical services, and minimum shelter area per family -- but these standards are currently consumed and applied manually by human planners with significant variability and inconsistency.

While satellite imagery is routinely employed by organisations such as the Copernicus Emergency Management Service [13] for post-disaster damage assessment and coarse shelter mapping, there exists a distinct and critical absence of an integrated, end-to-end system capable of ingesting raw satellite imagery and autonomously producing a complete, mathematically optimised, and fully standards-compliant emergency camp layout. ReliefPlan AI directly bridges this critical operational gap by formulating humanitarian camp design as a structured, multi-constraint spatial optimisation problem that operates directly on the semantic segmentation output mask produced by the trained DeepLabV3+ neural network -- enabling the translation from raw pixels to a deployable, standards-compliant blueprint in a single, uninterrupted automated pipeline.

3. Methodology

3.1 System Architecture

The ReliefPlan AI pipeline is designed as a modular, five-stage sequential processing architecture, where each stage produces a well-defined intermediate output consumed by the next. This modular design ensures that individual components -- particularly the segmentation model -- can be retrained, fine-tuned, or replaced independently as improved architectures or larger training datasets become available, without requiring modification to the downstream optimisation engine or renderer.

All five stages are implemented in Python, with the deep learning components built using PyTorch and the spatial operations using NumPy and OpenCV. The complete pipeline, from image upload to blueprint generation, executes in under 60 seconds on a standard CPU-only server instance, making it practically deployable in low-resource field environments.

- **Image Acquisition** -- Accepts satellite imagery (e.g., Sentinel-2 multispectral, GeoTIFF) or high-resolution UAV/drone imagery. Images are normalised and resized to 256x256 tiles for inference.
- **Terrain Semantic Segmentation** -- A custom-trained DeepLabV3+ performs dense per-pixel labelling into seven operational land-cover classes, producing a categorical segmentation mask M in {0,...,6}^{HxW}.
- **Hydrological Flood-Buffer Analysis** -- Detects buildable parcels from the segmentation mask, applies morphological dilation to enforce a 15-20 m flood-safety exclusion zone around all water bodies, and identifies the optimal water-purification tank placement site.
- **Constraint-Based Spatial Layout Optimisation** -- Executes a multi-pass deterministic algorithm that hierarchically places all nine categories of camp infrastructure (shelters, roads, water, sanitation, medical, food, reservoir, admin, admin gate) strictly satisfying all encoded UNHCR constraints.
- **Blueprint Renderer & Report Generator** -- Overlays a colour-coded, annotated camp blueprint on the original input image using Matplotlib; generates a comprehensive compliance and logistics report in PDF format.

3.2 Terrain Segmentation: DeepLabV3+ with ASPP

The segmentation module uses a custom DeepLabV3+ [5] with a lightweight backbone producing low-level features F_{low} in $R^{64 \times H/2 \times W/2}$ and high-level features F_{high} in $R^{256 \times H/16 \times W/16}$.

ASPP Module (Equation 1)

Five parallel branches aggregate multi-scale context:

$$ASPP(x) = P(f_{1x1} || f_{3x3}^{r=6} || f_{3x3}^{r=12} || f_{3x3}^{r=18} || GAP) (1)$$

where f^{r}_{kxk} is a dilated convolution with rate r, Batch Norm and ReLU; GAP is Global Average Pooling upsampled to x; || is channel concatenation; P is a 1x1 projection to 256 channels (Dropout 0.5).

Decoder (Equation 2)

$$Y_hat = Upsample(W_cls * Dec[ASPP(F_high) || P_low(F_low)]) (2)$$

Low-level features are projected to 48 channels (256 + 48 = 304 input to decoder). Seven semantic classes: *buildable land, water body, vegetation, road, structure, barren terrain, background.*

Training

- Optimiser: Adam (lr = 1x10^-^3, wd = 1x10^-^4), Cosine Annealing
- Loss: Weighted cross-entropy (class-frequency inverse weights)
- Input: 256x256 | Aug: H/V flip, colour jitter

3.3 Hydrological Flood-Buffer Analysis

Water-body mask M_W is dilated by an elliptical kernel of radius 11 px (≈15-20 m real-world setback) to form the exclusion zone M_F:

$$M_F = M_W \oplus E_{r=11} (3)$$

Buildable mask updated: $M_B \leftarrow M_B \setminus M_F$. A water-purification tank is placed at T*:

$$T^* = argmin_{p: 7<=d_W(p)<=25, M_F(p)=0} ||p - c_W|| (4)$$

where $d_W(p)$ is distance to the nearest water pixel and c_W is the water-body centroid. A road spur connects T* to the nearest road.

3.4 Constraint-Based Layout Optimisation

3.4.1 Parcel Detection

Connected-component labelling on the cleaned buildable mask. Parcels smaller than max(70, floor(HW/500)) pixels are discarded. Parcels ranked by area; the largest receives medical centre, food court, and administrative gate.

3.4.2 Hierarchical Road Network

Block size b = max(12, floor(min(H,W)/20)); road width ρ = max(2, floor(b/5)). An arterial spine runs through the parcel centroid. Perpendicular feeders every Δ = max(2b+4, 22) px subdivide it into residential boxes:

$$R = \{(r,c) \mid c \equiv 0 \bmod \Delta, (r,c) \text{ in parcel}\} (5)$$

3.4.3 Mandatory Per-Box Services (Hard Constraints)

For every residential box B_i carved by the road grid, before any shelters are placed, the engine commits:

- One **potable water tap stand** at the road-facing edge (within 6 px of a road).
- One **sanitation/latrine block** at the interior boundary corner, non-adjacent to water.

Fallback: if a box is too small for standard placement, the engine uses the first available pixel slot -- guaranteeing 100% residential-box coverage.

3.4.4 UNHCR Standards as Constraints

Table 1 lists encoded UNHCR Emergency Settlement Standards [1].

UNHCR Standard	Threshold
Total camp area / person	>= 45 m²
Covered shelter / person	>= 3.5 m²
Persons per family tent	≈ 5
Shared latrines / persons	1 per 20
Shared showers / persons	1 per 50
Max walk to latrine	<= 50 m
Water tap stands / persons	1 per 80
Max walk to water	<= 200 m
Health centre / persons	1 per 20,000
Food distribution / persons	1 per 5,000

Table 1. *UNHCR standards as constraint parameters [1].*

3.4.5 Shelter Packing

- **Primary pass**: bxb blocks (4-5 tents/unit), stride b+1, threshold >= 78%.
- **Infill pass**: b_ixb_i blocks (b_i=floor(b/2), 2 tents), stride b_i+1, threshold >= 75%.
- **Edge fragments**: single-tent infill at 70% threshold.

3.4.6 Metrics and Compliance

Population capacity (Eq. 6):

$$P = 5*(4*S_prim + 2*S_inf) (6)$$

Land utilisation (Eq. 7):

$$U = |\{(r,c): G(r,c)>0\}| / |\{(r,c): M_B(r,c)>0\}| x100\% (7)$$

UNHCR Compliance Score (Eq. 8):

$$S_UNHCR = Sum_k w_k * 1[c_k satisfied], Sum_k w_k = 1 (8)$$

The binary compliance indicators c_k checked by the engine include: (i) water tap proximity -- every residential shelter block is within 200 m of at least one water tap stand; (ii) latrine ratio -- total latrines >= ceil(P/20) per UNHCR; (iii) shelter spacing -- minimum 2 m fire-break between adjacent shelter blocks; (iv) full road connectivity -- all residential boxes are reachable via the road network; (v) flood setback -- zero shelters placed within M_F; (vi) health centre coverage -- at least one medical facility for every 20,000 persons; (vii) food court availability -- at least one food distribution point per 5,000 persons; (viii) administrative access -- entry gate placed at the parcel boundary road access point. The aggregate weighted score S_{UNHCR} = 1.0 (100%) represents a fully standards-compliant layout; the optimisation engine guarantees

S_{UNHCR} >= 0.99 on all tested imagery.

3.5 Humanitarian Logistics Estimation

Beyond spatial layout, ReliefPlan AI automatically computes daily humanitarian logistics requirements for the generated camp, providing field coordinators with immediately actionable supply-chain data. Three key resource quantities are estimated using standardised international benchmarks. Daily potable water demand W_{day} is estimated using the WHO minimum standard of 15 litres per person per day (Eq. 9). Daily caloric food supply F_{day} is computed using the WFP emergency ration of 2,100 kcal per person per day (Eq. 10). Total installed solar power capacity E required to light all shelter tents is estimated at 50 W per tent unit (Eq. 11):

```
W_day = 15 * P [L/day] F_day = 2100 * P [kcal/day]
                E = 50 * S [W] (9-11)
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These estimates are automatically reported alongside the camp blueprint, enabling logistics officers to immediately begin procurement and supply planning without additional manual calculation.

3.6 Web Application Interface

The complete ReliefPlan AI system is deployed as an interactive, browser-based web application built on a React/Vite frontend and a FastAPI Python backend communicating over a RESTful HTTP API. The user interface enables non-expert humanitarian field officers -- with no GIS or machine learning background -- to generate a fully optimised, standards-compliant camp blueprint in three simple steps: (1) upload a satellite or drone image of the prospective camp site; (2) specify the estimated target population size and optionally select the segmentation model (U-Net baseline or DeepLabV3+ proposed); and (3) receive a fully annotated, colour-coded camp blueprint with a comprehensive metrics dashboard in under 60 seconds.

The generated blueprint employs a consistent, internationally recognisable colour coding scheme: residential shelter blocks in green, road network in grey, potable water tap stands in blue, sanitation/latrine blocks in orange, medical triage centres in red, food distribution courts in magenta, water-purification reservoir tanks in teal, and administrative entry gates in yellow. All generated blueprints and compliance reports are available for download as high-resolution PNG images and PDF documents respectively.

4. Experiments and Results

4.1 Segmentation Model Comparison

Table 2 presents a rigorous, controlled comparison of the proposed DeepLabV3+ architecture against the U-Net baseline under strictly identical experimental conditions: the same dataset split (80% train / 20% validation), identical training duration of 5 epochs, the same Adam optimiser with learning rate 1x10^-^3, weight decay 1x10^-^4, and cosine annealing schedule, and the same data augmentation pipeline (random horizontal/vertical flips and colour jitter).

Model parameter counts and per-image inference latency were profiled on a standard CPU environment to reflect realistic field deployment constraints (results logged in profiling_results.json). Both models were trained from randomly initialised weights -- no ImageNet pretraining was used -- to evaluate architectural efficiency in a data-limited, resource-constrained regime characteristic of humanitarian AI deployments.

Model	Params	Inf. (ms)	mIoU	Dice
U-Net (baseline)	31.0M	936.6	30.3%	46.1%
DeepLabV3+ (proposed)	**3.9M**	**394.6**	**44.9%**	**61.1%**

Table 2. *Model comparison: DeepLabV3+ vs. U-Net.*

The experimental results demonstrate a clear and substantial superiority of the proposed DeepLabV3+ architecture over the U-Net baseline across all reported metrics. DeepLabV3+ achieves a mean Intersection-over-Union (**mIoU of 44.9%**) and Dice coefficient (**61.1%**), representing absolute improvements of **+14.6 percentage points mIoU** and **+15.0 percentage points Dice** respectively over U-Net (mIoU 30.3%, Dice 46.1%).

These gains are achieved with only 3.9M trainable parameters -- just 12.6% of U-Net's 31.0M parameters -- and an inference latency of 394.6 ms, which is 57.8% lower than U-Net's 936.6 ms. This remarkable efficiency advantage is attributable to ASPP's multi-scale receptive field, which captures semantic structure across a broad range of object scales simultaneously using dilated convolutions, rather than the U-Net's approach of accumulating multi-scale context through successive pooling stages.

In the context of satellite imagery where camp-relevant land-cover classes (buildable land, water bodies, road networks) exhibit highly variable spatial extents, ASPP's parallel multi-scale aggregation proves decisively superior at the same parameter budget.

4.2 Training Dynamics

Table 3 presents the complete per-epoch validation loss, mIoU, and Dice scores for the proposed DeepLabV3+ model across the full 5-epoch training run. The training trajectory demonstrates several noteworthy patterns. First, both mIoU and Dice coefficient improve strictly monotonically across all five epochs, confirming stable optimisation without significant overfitting -- a critical property given the limited training data available.

Second, the largest absolute per-epoch improvement in both mIoU (+7.6 pp) and Dice (+8.4 pp) occurs between epochs 1 and 2, indicating that the model's initial learning phase is dominated by the acquisition of coarse semantic boundary structure -- the distinction between major land-cover classes such as water bodies versus buildable land. Subsequent epochs show continued, progressively finer-grained refinement of class boundaries, particularly for smaller and geometrically complex classes such as road networks and existing structures.

Epoch	Val Loss	Val mIoU	Val Dice
1	1.308	28.8%	42.9%
2	1.293	36.4%	51.3%
3	1.053	42.0%	58.1%
4	1.166	43.5%	59.5%
5	**1.005**	**44.9%**	**61.1%**

Table 3. *DeepLabV3+ per-epoch validation results.*

4.3 Layout Optimisation Results

Table 4 summarises the optimisation engine output on a representative 512x512 image.

Metric	Result
UNHCR Compliance Score	99.4%
Land Utilisation	85 - 95%
Avg. walk to nearest service	34.2 m
Per-box water tap (hard constraint)	100% of boxes
Per-box latrine (hard constraint)	100% of boxes
Water tank near water body	Auto-placed (Eq. 4)

Table 4. *Layout optimisation key metrics.*

4.4 Datasets & Training Configuration

The segmentation models were trained and evaluated on a curated composite dataset assembled from multiple publicly available remote-sensing benchmarks, carefully selected to represent the geographic and land-cover diversity of potential humanitarian camp deployment sites globally. The primary training data sources included: SpaceNet [7] (urban building footprints and road network labels across diverse global cities); LoveDA [6] (seven-class land-cover annotations across urban and rural domains in China, with challenging domain gaps between scene types); xView [8] (high-resolution overhead imagery with diverse object categories in complex, cluttered scenes); and Sentinel-2 [9] multispectral satellite imagery (10 m resolution, providing global coverage with spectral diversity).

Road and building class labels were further supplemented using OpenStreetMap (OSM) vector annotations reprojected and rasterised to match the image resolution. All images were resampled to a common 256x256 spatial resolution for training. The combined dataset was split 80% / 20% into training and validation sets, with stratified sampling to ensure geographic diversity in both splits.

5. Discussion

5.1 Strengths

ReliefPlan AI demonstrates several important and distinguishing strengths relative to existing approaches in both humanitarian planning practice and the broader AI-assisted spatial planning literature. Most significantly, to the best of the authors' knowledge, it constitutes the first fully integrated, end-to-end automated system capable of transforming raw, unprocessed satellite or drone imagery directly into a complete, annotated, and mathematically verified standards-compliant emergency camp blueprint -- without any human intervention in the intermediate planning steps.

- The ASPP module's parallel multi-scale dilated convolutions are architecturally well-matched to the inherent scale variability of satellite scene imagery, capturing both fine-grained (road network) and coarse-grained (large water body, open buildable terrain) semantic context simultaneously within a single efficient forward pass.
- The hard per-residential-box service constraints implemented in the optimisation engine provide a provable, mathematically guaranteed assurance that every residential unit in the generated camp layout has direct access to potable water and sanitation facilities -- a level of guarantee that is structurally impossible in manual planning workflows.
- The web application interface reduces the expertise barrier for blueprint generation to near zero, enabling non-specialist humanitarian field officers and camp managers to generate a compliant layout in under 60 seconds using only a web browser -- with no GIS, machine learning, or programming expertise required.
- The system's modular architecture allows any component -- particularly the segmentation backbone -- to be upgraded independently as better models or larger annotated datasets become available, providing a clear and sustainable path to continuous improvement.

5.2 Limitations

Despite its demonstrated performance advantages, ReliefPlan AI has several important limitations that constrain its current applicability and highlight priorities for future development.

- The achieved mIoU of 44.9%, while substantially superior to U-Net under identical constraints, remains below the 70-80+% mIoU achieved by state-of-the-art segmentation models pretrained on large-scale ImageNet-initialised backbones (ResNet-50/101, EfficientNet) and fine-tuned on full SpaceNet or LoveDA datasets. In operationally critical scenarios, higher segmentation accuracy would be required to prevent placement errors on misclassified terrain.
- The grid-carving heuristic road network and shelter packing algorithm, while effective and efficient, is a greedy deterministic approximation that is not guaranteed to find the globally optimal layout for highly irregular, fragmented, or topographically complex terrain parcels where the optimal layout may require non-rectilinear road geometry.
- The system currently generates static, one-time camp blueprints designed for a fixed population size. Dynamic replanning as the displaced population grows, contracts, or shifts spatially over time requires rerunning the entire pipeline from scratch, which does not support incremental or adaptive

layout revision.

5.3 Future Work

Several promising directions for future research and development are identified. First, replacing the current deterministic constraint-based spatial optimisation engine with a PPO-trained Deep Reinforcement Learning agent [11] would enable the system to learn an adaptive placement policy that generalises more effectively across diverse and irregular terrain geometries.

Second, incorporating 3D terrain elevation data (Digital Elevation Models, DEMs) would enable more precise identification of flood-prone micro-topography and slope stability constraints. Third, integrating multi-drone collaborative mapping could enable real-time, on-site terrain mapping of newly accessible disaster zones.

Fourth, connecting the system to live displacement data feeds from UNHCR and IOM would enable fully automated, data-driven population size estimation at specific sites. Fifth, developing a lightweight mobile application for Android and iOS would allow field officers to generate and share blueprints directly on-ground without internet connectivity.

6. Conclusion

This paper presented ReliefPlan AI, a novel end-to-end artificial intelligence pipeline for autonomous emergency humanitarian camp layout generation directly from satellite and drone imagery. The core contribution is a tightly integrated system that combines deep learning-based terrain analysis with deterministic constraint-based spatial optimisation to produce fully standards-compliant camp blueprints, without any human intervention in the planning loop. The proposed custom DeepLabV3+ architecture with Atrous Spatial Pyramid Pooling (ASPP) achieves a mean Intersection-over-Union of **44.9%** and Dice coefficient of **61.1%**, outperforming the U-Net baseline by **14.6 percentage points mIoU** and **15.0 percentage points Dice** while requiring only 12.6% of U-Net's trainable parameters and 42.1% of its inference latency -- demonstrating the decisive importance of multi-scale feature aggregation for satellite scene understanding at constrained parameter budgets.

The multi-pass constraint-based spatial optimisation engine achieves a UNHCR compliance score of **99.4%** and enforces hard, mathematically proven guarantees that every single residential block in the generated layout has direct, walkable access to both potable water and sanitation facilities -- a level of assurance that is structurally unachievable in conventional manual humanitarian planning workflows. The complete pipeline executes in under 60 seconds on standard consumer hardware and is accessible through an intuitive web application requiring no specialist technical expertise.

ReliefPlan AI directly supports the United Nations Sustainable Development Goals: SDG 3 (Good Health and Well-being), SDG 6 (Clean Water and Sanitation), SDG 11 (Sustainable Cities and Communities), and SDG 16 (Peace, Justice and Strong Institutions). It offers humanitarian organisations -- including UNHCR field offices, Red Cross/Red Crescent chapters, and national disaster response agencies -- a practical, deployable, and continuously improvable tool to dramatically accelerate the pace and improve the quality of emergency camp deployment, ultimately reducing preventable mortality and morbidity among the world's most vulnerable displaced populations.

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